

H1 / H2 / H3 Matrix Report

Source: results.csv · cells: 68 · targets: 17 · conditions: full, no-fusion, no-replan, random-order

Executive Summary

Total cells	68
Distinct targets	17
Mean duration / cell	169.1s
Total findings across cells	18428
Engine pool example	nmap,url_crawler,http_probe,nmap_vuln,nikto,header_vuln_check

Findings per (target × condition)

Target	full	no-fusion	no-replan	random-order
aloro	358.0	39.0	14.0	354.0
bodgeit	343.0	28.0	14.0	343.0
bwapp	350.0	33.0	14.0	349.0
commix_testbed	353.0	39.0	14.0	357.0
dvwa_live	2347.0	2033.0	14.0	2347.0
juiceshop	358.0	44.0	14.0	361.0
nowasp	356.0	38.0	14.0	352.0
pygoat	342.0	23.0	14.0	342.0
skf_csrf	340.0	26.0	14.0	340.0
skf_idor	344.0	26.0	14.0	346.0
skf_jwt_null	343.0	26.0	14.0	342.0
skf_lfi	340.0	26.0	14.0	340.0
skf_rfi	344.0	26.0	14.0	340.0
skf_url_redirection	344.0	26.0	14.0	238.0
skf_xss	343.0	26.0	14.0	340.0
vampi	345.0	31.0	14.0	345.0
webgoat	342.0	28.0	14.0	344.0

H1 — Closed-loop replanning improves coverage

mean(full)	464.2
mean(no-replan)	14.0
paired t (all)	t=3.826, p=0.0007
Wilcoxon (all)	W=153.0, p=0.000143
paired t (no DVWA)	t=211.759, p=0.000000, Cohen d=52.940

mean lift (no DVWA)	+2375%
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H2 — Cross-tool fusion

mean(full)	464.2
mean(no-fusion)	148.1
paired t	t=642.890, p=0.0000
Wilcoxon	W=153.0, p=0.000127

H3 — Decision-quality stratification

Condition	Correct	Suboptimal	Incorrect	Total	% incorrect
full	102	0	34	136	25.0%
no-fusion	85	17	34	136	25.0%
no-replan	51	0	0	51	0.0%
random-order	102	0	17	119	14.3%